

An Introduction to the yocto PROJECT



The Yocto Project is an open source project initiated by the Linux Foundation, which makes development on embedded devices easier and portable. It's an end-to-end embedded Linux development environment with various tools, configuration files and documentation, and pretty much has everything one needs. This article attempts to introduce the project, and is by no means a complete tutorial. It assumes that the reader is familiar with the nuances of embedded development environments.

The industry has seen a proliferation of embedded devices and processors. As these have grown more powerful and feature-rich, the use and popularity of the Linux operating system has skyrocketed. One of the driving factors for its popularity is because it remains open source, offers free licensing and extensive usage, which in turn has led to an abundance of applications.

However, embedded developers cannot simply pick up a standard Linux distribution or application for use in their tiny environments. They face challenging tasks like handpicking boot-loaders, the kernel, libraries, applications, and development tools specific to their custom hardware environment, before cross-compiling and optimising them to fit in their minuscule embedded environments.

The solutions to these challenges are not easy, and a developer's group will need considerable effort, experience and exposure to address them. Moreover, spending time on these problems will leave developers with little time and energy to really concentrate on the core task of building excellent features for their boards. Non-commercial and commercial embedded Linux distributors offer well-tested solutions for specific embedded processors, but for an embedded developer, this is a nightmare. Every new project demands the learning of a new set of tools and development environments prior to even getting started.

Evolution

It all started with the OpenZaurus distribution, which had a tool called *buildroot* to solve some of the problems faced by embedded developers. The constraints with *buildroot* were that it could not scale to different targets apart from Sharp's Zaurus. Its users eventually began discussing about the possibilities of a new generic build, which led to the beginning of OpenEmbedded (OE). This is an embedded build system, and provides a 'build from source' methodology, to reliably bootstrap customised Linux distributions that cater to embedded devices and beyond. It has a robust structure for metadata, and vast coverage for open source software. Feature

wise, it came out as the most advanced system in the market, and being specifically designed for the customisation needs of the embedded market space, soon became the industry standard. A large number of organisations were using OE, which in turn increased support and enhanced the internals of the OE. While being a very good first-of-its-kind base, OE suffered due to its complexity, steep learning curve and lack of documentation. Apart from being a build system, OE started to collate various other tools as well—while it was inherently not designed to do so.

In November 2010, industry leaders joined together to form the Yocto Project under the umbrella of the Linux Foundation, and announced that the work would continue under this new name. Despite Yocto being a fork of OpenEmbedded, the common areas are now down to OE-core. This is mutually beneficial for both OE and Yocto, where the resultant output is a high-quality OE-core with fewer fully featured software tools in a single framework. Yocto got a head start with a project built on commercial learnings and community acceptance.

The Yocto Project

According to the Yocto Project website: "It's not an embedded Linux distribution—it creates a custom one for you."

With its vibrant community, the Yocto Project has rapidly gained traction among silicon companies, embedded developers, operating system providers and tools providers, thus capturing the entire gamut of embedded development. Some of the famous names involved are Cavium Networks, Dell, Freescale Semiconductor, Intel, Texas Instruments, etc. Despite being funded by several major organisations, the Yocto Project remains independently governed by the Linux Foundation, based on the open source principles of collaboration and transparency.

Yocto gives developers a head start by providing hardware BSPs (Board Support Packages) a standardised format across architectures, the latest and well-tested metadata for the kernel, an 'automagically' created Application Developer Kit (ADK), and great flexibility for embedded device requirements.

As per Yocto's documentation page (yoctoproject.org/)